

Cost Analysis

2026-05-25

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1 Assumptions

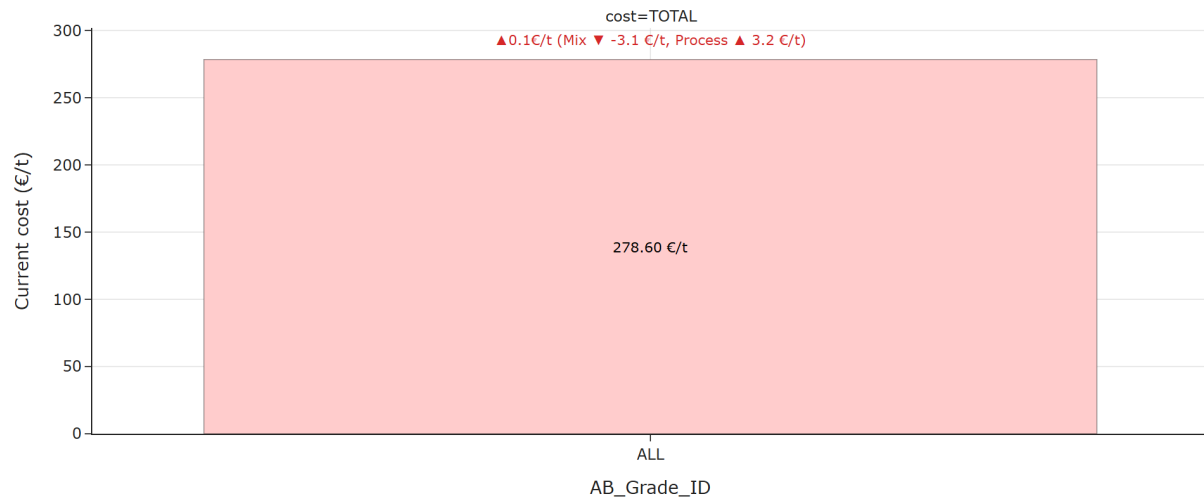
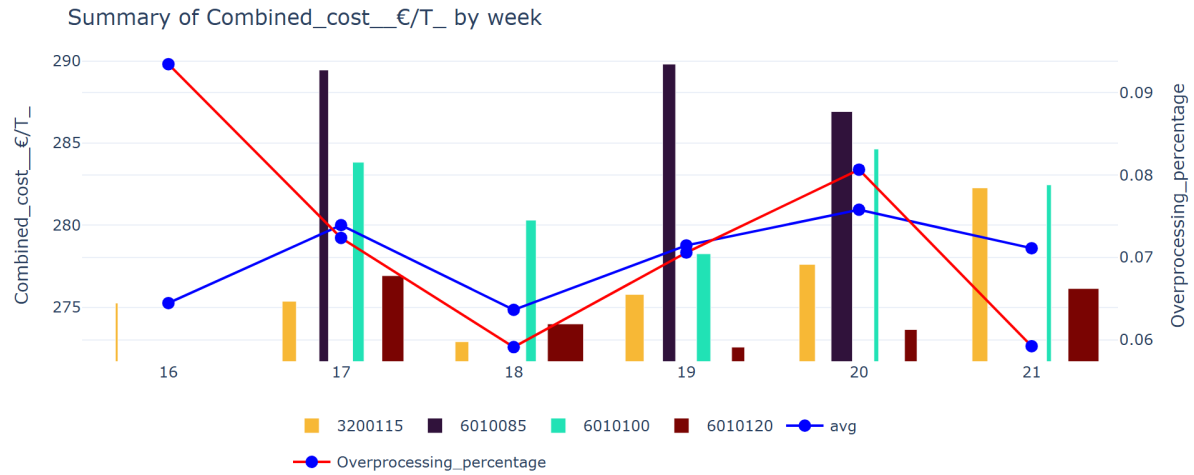
The aim of the analysis is to identify the most cost-effective production periods in months of data and recommend optimal operating windows for the corresponding key influencers that can be set as a benchmark for the weekly analysis.

- **Period Analyzed:** April 19, 2026 – May 25, 2026
- **Cost Components Included:** Fibre, steam, electricity, starch, and chemicals.
- **Grades Included:** Only grades 6010085, 6010100, 601020 and 3200115 are included in this week’s analysis.
- **Data Selection Criteria:** Only operating conditions where strength properties exceed specification thresholds were considered.
- **Naming Conventions Used:**
 - **Current:** Refers to the last week of the analyzed period.
 - **Historic:** Represents the average cost over the monthly interval previous to the “Current” period. Used as baseline.
 - **Best:** Corresponds to the cost associated with the optimal batch configuration.

2 Executive Summary

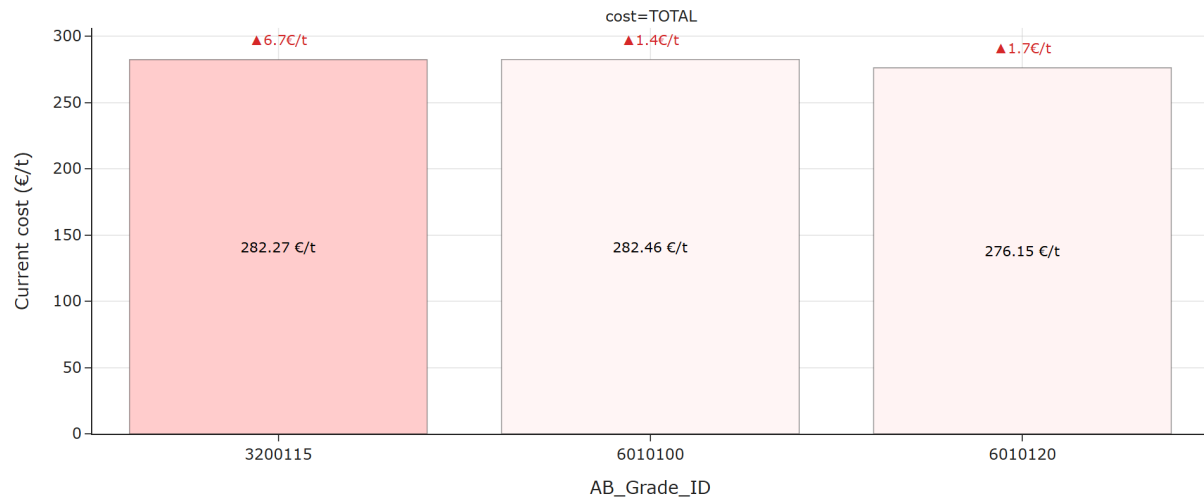
The following plot illustrates how total cost varies across different grades, enabling a comparison between the current configuration and the historical baseline.

The next plot displays how total cost evolves over time, with a breakdown by grade to show individual contributions.



For all considered grades, TOTAL cost saw an increase of 0.07 €/t, resulting in 278.60 €/t.

Breaking down the overall cost change, -3.12 €/t was driven by a cheaper mix of grades, and +3.18 €/t was due to a deterioration in the process.

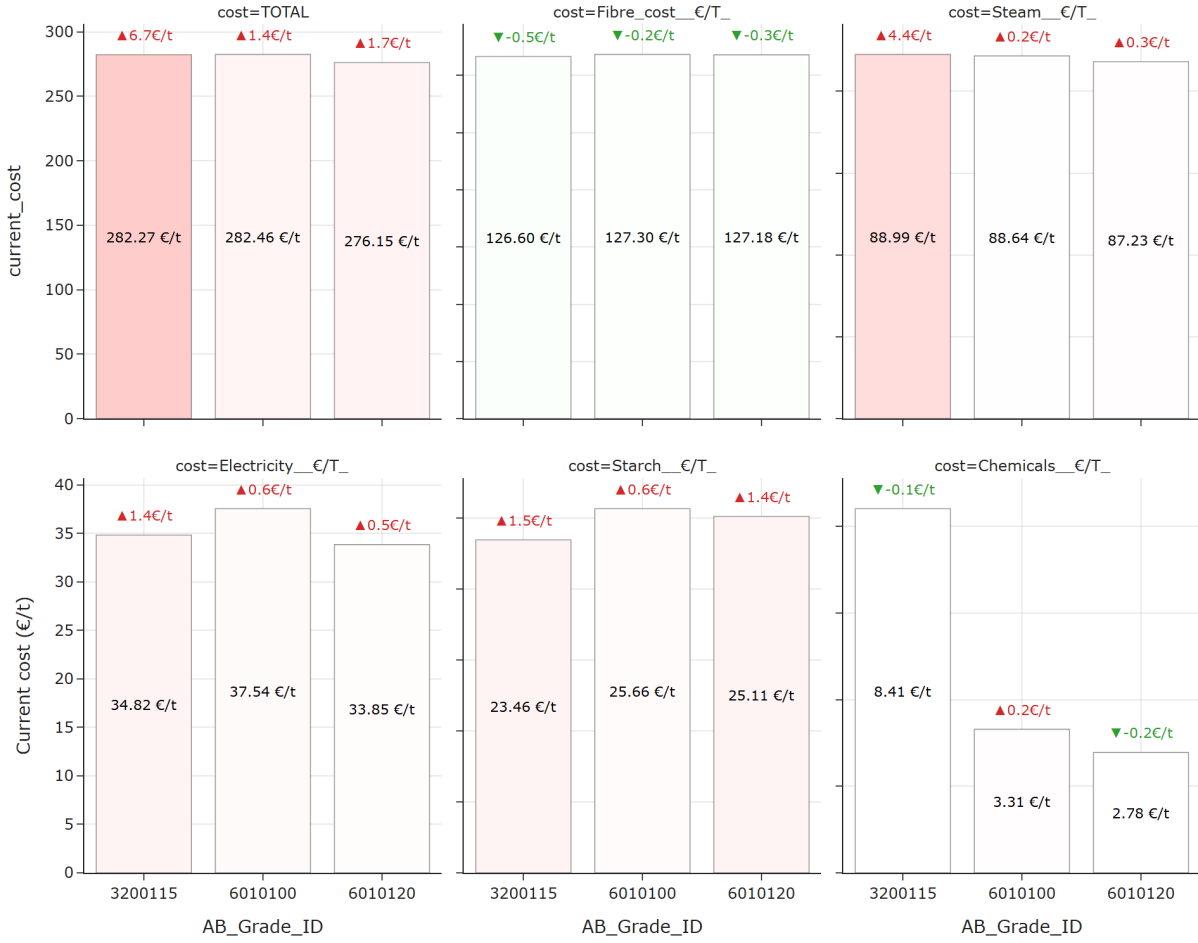


For grade 3200115, TOTAL cost saw an increase of 6.74 €/t, resulting in 282.27 €/t.

For grade 6010100, TOTAL cost saw an increase of 1.36 €/t, resulting in 282.46 €/t.

For grade 6010120, TOTAL cost saw an increase of 1.65 €/t, resulting in 276.15 €/t.

Looking across individual grades, the strongest cost increase was observed for grade 3200115 (+6.74 €/t). These movements stand out and may require further investigation.



Average change by component:

- **Steam_€/T** increased on average (+1.63 €/t).
- **Starch_€/T** increased on average (+1.15 €/t).
- **Electricity_€/T** increased on average (+0.83 €/t).
- **Fibre_cost_€/T** decreased on average (-0.35 €/t).
- **Chemicals_€/T** decreased on average (-0.01 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Steam_€/T** (+1.63 €/t on average), with the highest increase in grade 3200115 (+4.43 €/t) and the largest average decrease comes from **Fibre_cost_€/T** (-0.35 €/t on average), with the biggest drop in grade 3200115 (-0.49 €/t). These components stand out and may require further investigation.



Average change by steam:

- **Steam_Afterdryers** increased on average (+0.97 €/t).
- **Steam_Whitewater** increased on average (+0.80 €/t).
- **Steam_Predryers** decreased on average (-0.23 €/t).
- **Steam_Heatexchangers** increased on average (+0.09 €/t).
- **Steam_Starchkitchen** increased on average (+0.01 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Steam_Afterdryers** (+0.97 €/t on average), with the highest increase in grade 3200115 (+2.53 €/t) and the largest average decrease comes from **Steam_Predryers** (-0.23 €/t on average), with the biggest drop in grade 6010120 (-1.12 €/t). These components stand out and may require further investigation.



Average change by electricity:

- **Electricity_Other** increased on average (+0.54 €/t).
- **Electricity_Specific** increased on average (+0.14 €/t).
- **Electricity_Wiresection** increased on average (+0.13 €/t).
- **Electricity_Press_and_Steambox** decreased on average (-0.07 €/t).
- **Electricity_Freqconverters** increased on average (+0.07 €/t).
- **Electricity_Afterdryer** increased on average (+0.05 €/t).
- **Electricity_Predryer** decreased on average (-0.03 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Electricity_Other** (+0.54 €/t on average), with the highest increase in grade 3200115 (+0.79 €/t) and the largest average decrease comes from **Electricity_Press_and_Steambox** (-0.07

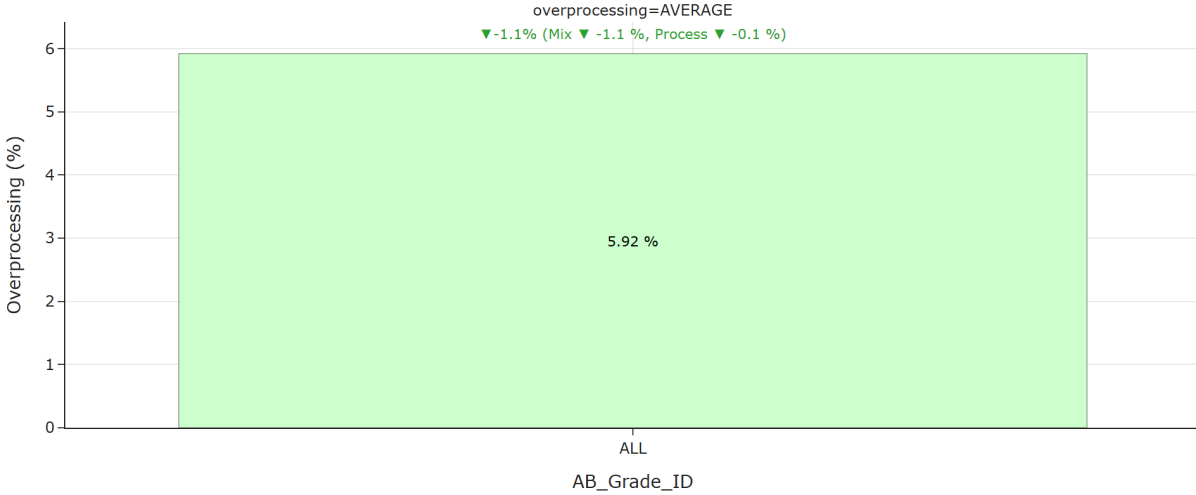
€/t on average), with the biggest drop in grade 3200115 (-0.14 €/t). These components stand out and may require further investigation.



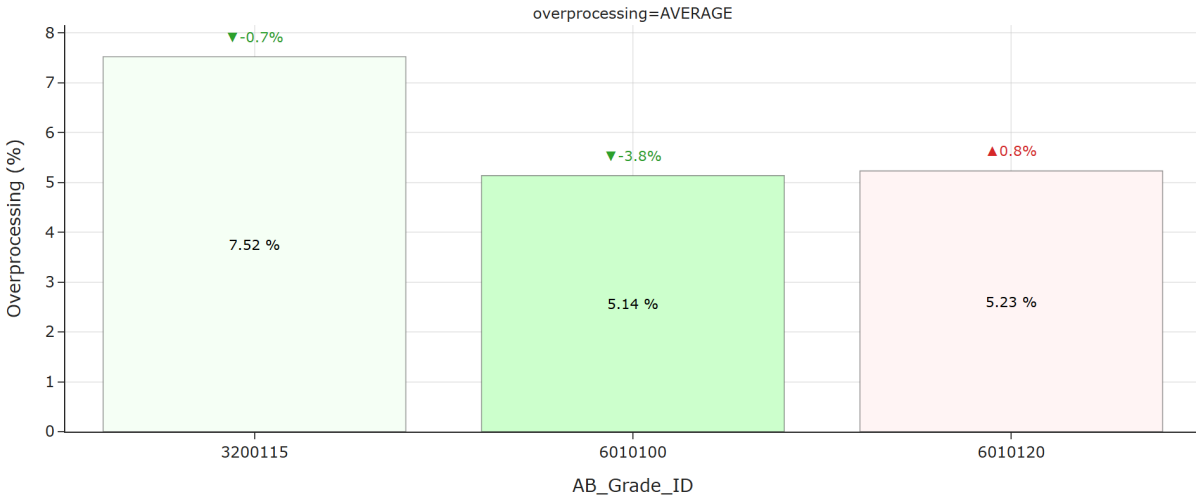
Average change by chemicals:

- **Fixative_2_€/T** increased on average (+0.19 €/t).
- **Defoamer_€/T** decreased on average (-0.07 €/t).
- **Bentonite_2_€/T** decreased on average (-0.04 €/t).
- **Sizing_Agent_€/T** increased on average (+0.02 €/t).
- **Retention_Aid_€/T** increased on average (+0.01 €/t).
- **Bentonite_1_€/T** decreased on average (-0.00 €/t).
- **Deaerator_€/T** increased on average (+0.00 €/t).
- **Natriumhydroxide_€/T** decreased on average (-0.00 €/t).

A more detailed breakdown by component shows that the largest average increase comes from **Fixative_2_€/*T*** (+0.19 €/t on average), with the highest increase in grade 6010100 (+0.48 €/t) and the largest average decrease comes from **Defoamer_€/*T*** (-0.07 €/t on average), with the biggest drop in grade 6010120 (-0.10 €/t). These components stand out and may require further investigation.



For all considered grades, TOTAL overprocessing saw a decrease of 1.15 %, resulting in 5.92 %.



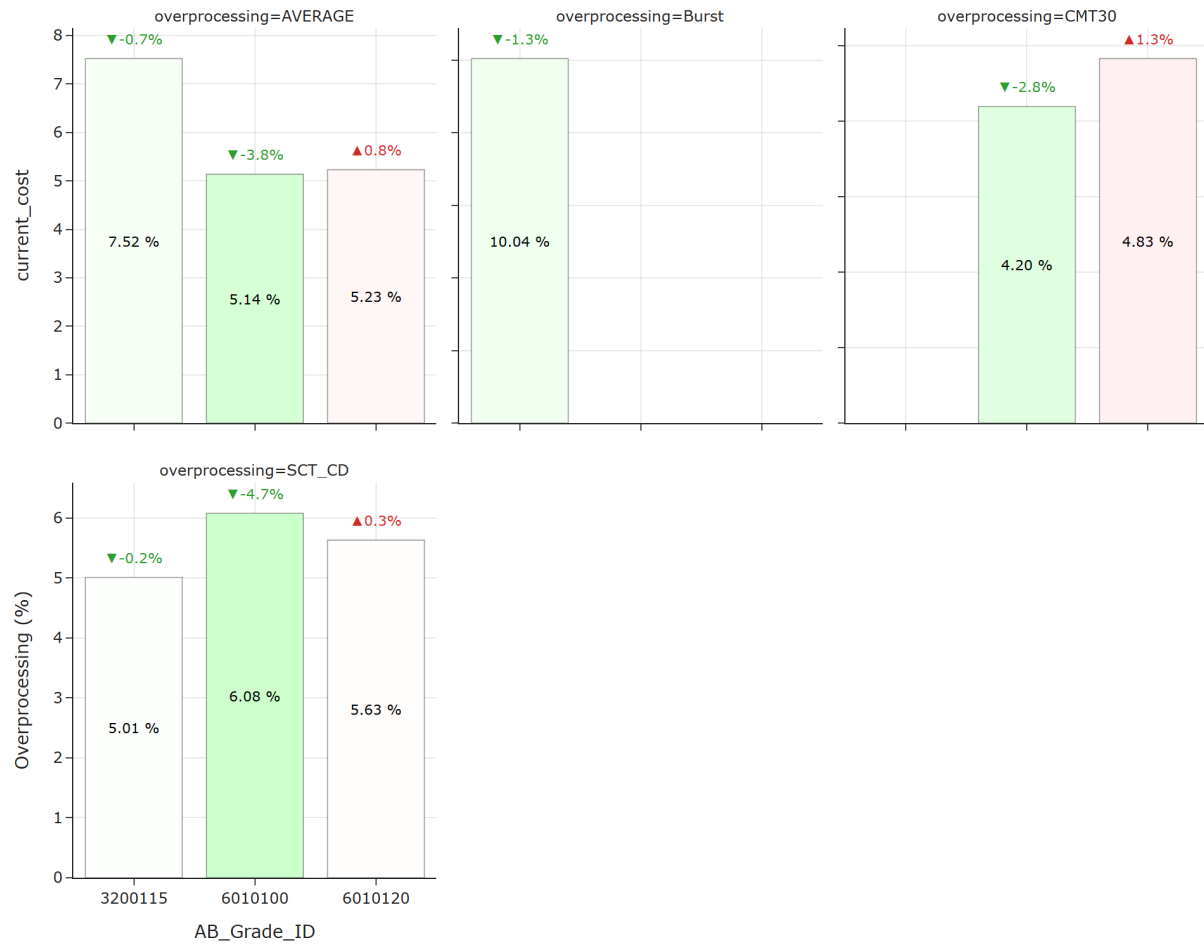
For grade 3200115, TOTAL overprocessing saw a decrease of 0.74 %, resulting in 7.52 %.

For grade 6010100, TOTAL overprocessing saw a decrease of 3.78 %, resulting in 5.14 %.

For grade 6010120, TOTAL overprocessing saw an increase of 0.82 %, resulting in 5.23 %.

Looking across individual grades, the strongest overprocessing increase was observed for grade 6010120 (+0.82 %) and the strongest overprocessing decrease was observed for grade 6010100 (-3.78 %). These movements stand out and may require further investigation.


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## AttributeError: 'DataFrame' object has no attribute 'cost'. Did you mean: 'count'?
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For grade 3200115, TOTAL overprocessing saw a decrease of 0.74 %, resulting in 7.52 %.

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For grade 6010120, TOTAL overprocessing saw an increase of 0.82 %, resulting in 5.23 %.

Looking across individual grades, the strongest overprocessing increase was observed for grade 6010120 (+0.82 %) and the strongest overprocessing decrease was observed for grade 6010100 (-3.78 %). These movements stand out and may require further investigation.

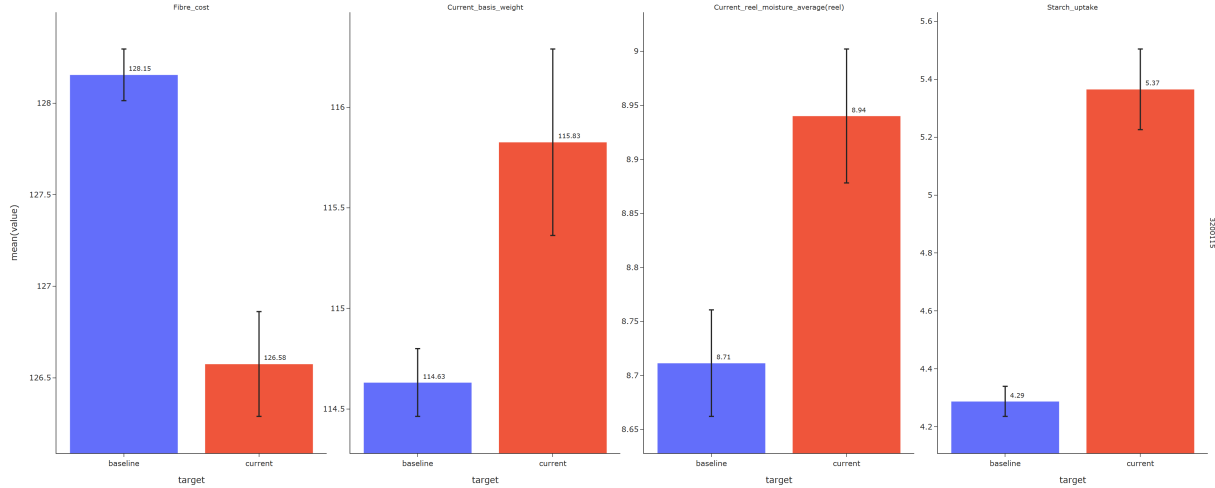
3 Breakdown by grades

3.1 3200115

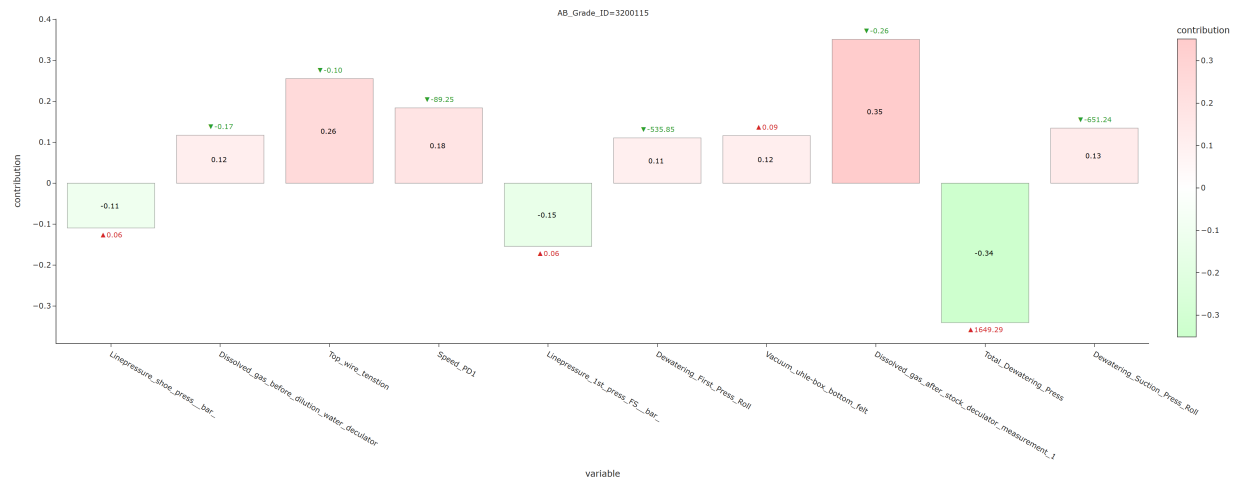
3.1.1 Fiber cost



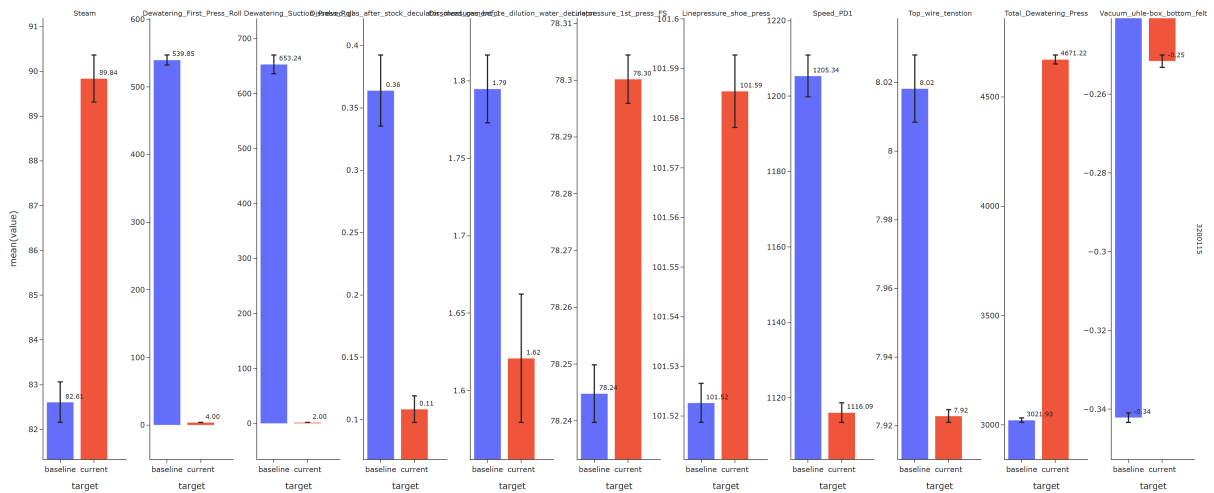
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Starch uptake g/m2 increased (+1.08)



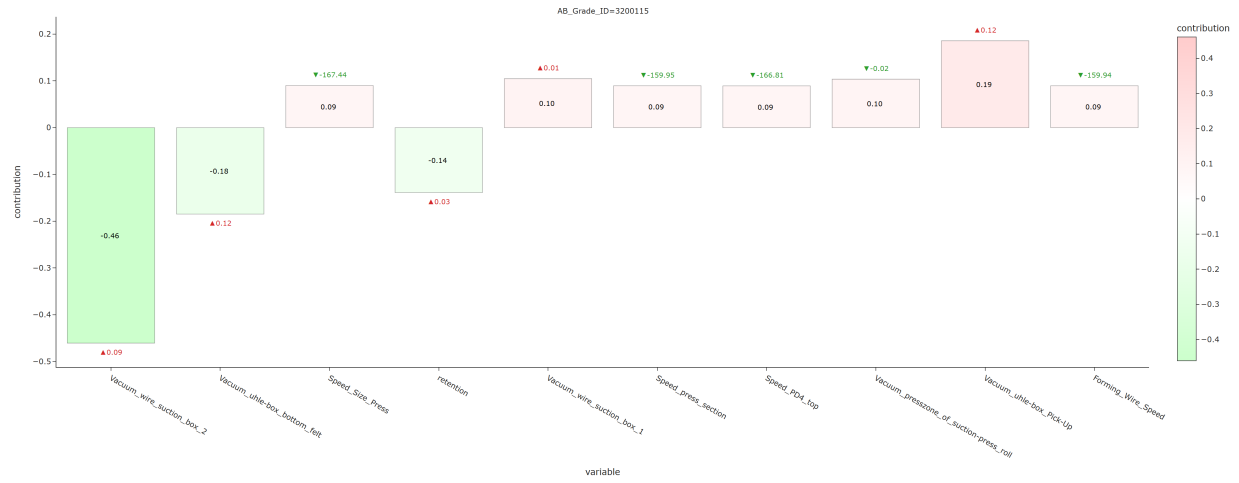
3.1.2 Steam cost



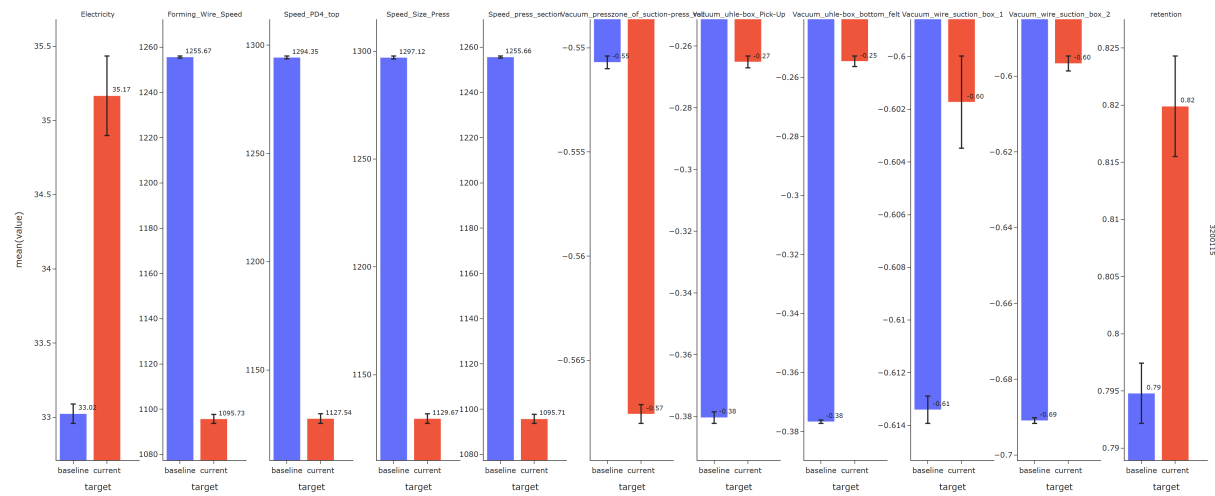
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Total Dewatering Press increased (+1649.29) - Linepressure 1st press FS bar increased (+0.06) Cost-increasing drivers: - Dissolved gas after stock decuator measurement 1 decreased (-0.26) - Top wire tension decreased (-0.10) - Speed PD1 decreased (-89.25) - Dewatering Suction Press Roll decreased (-651.24) - Dissolved gas before dilution water decuator decreased (-0.17) - Vacuum uhle-box bottom felt increased (+0.09)



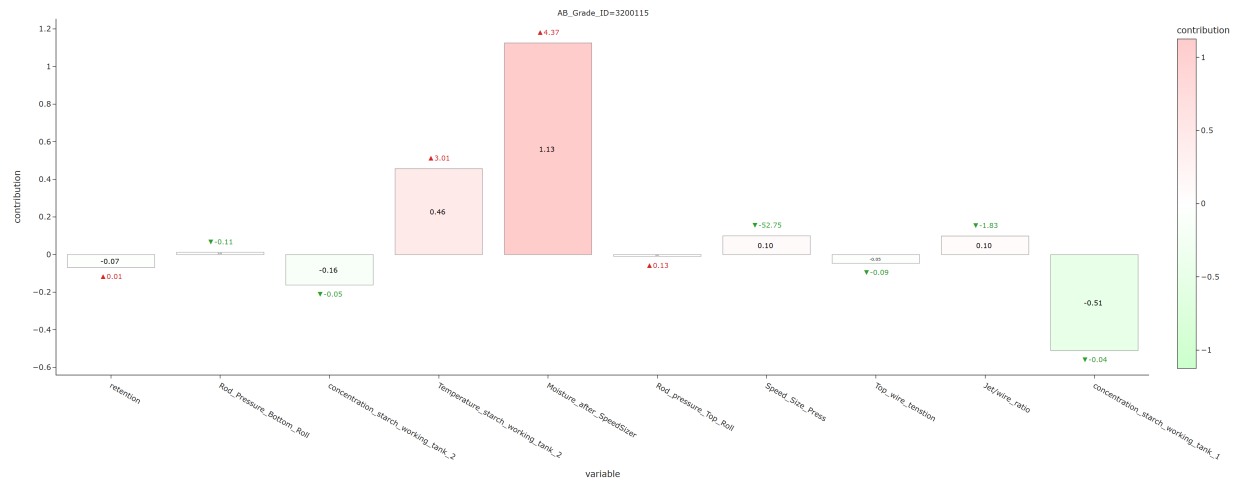
3.1.3 Electricity cost



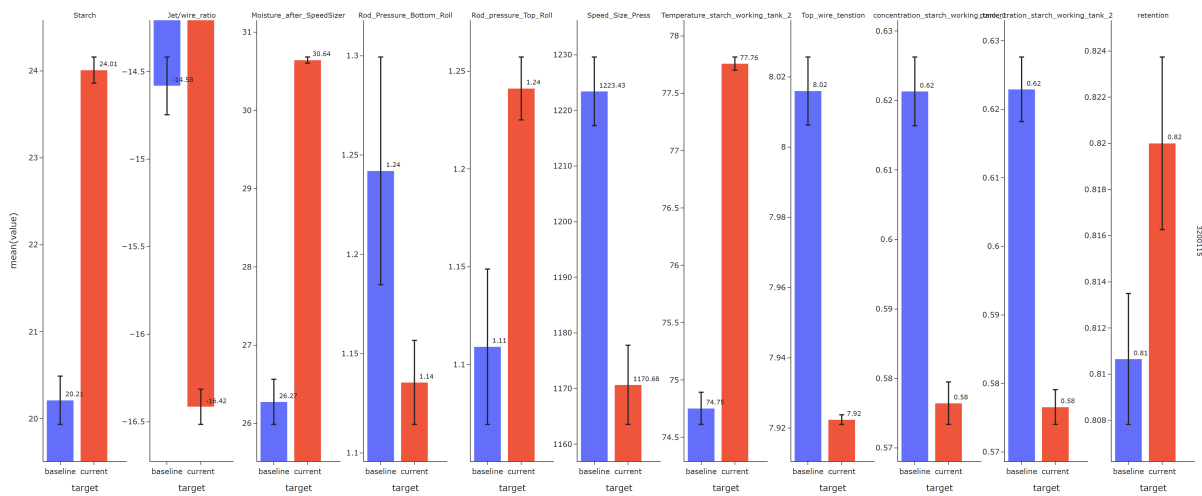
For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Vacuum wire suction box 2 increased (+0.09) - Vacuum uhle-box bottom felt increased (+0.12) - Retention increased (+0.03) Cost-increasing drivers: - Vacuum uhle-box Pick-Up increased (+0.12) - Vacuum wire suction box 1 increased (+0.01) - Vacuum presszone of suction-roll decreased (-0.02)



3.1.4 Starch cost



For grade 3200115, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Concentration starch working tank 1 decreased (-0.04) Cost-increasing drivers: - Moisture after SpeedSizer increased (+4.37) - Temperature starch working tank 2 increased (+3.01)

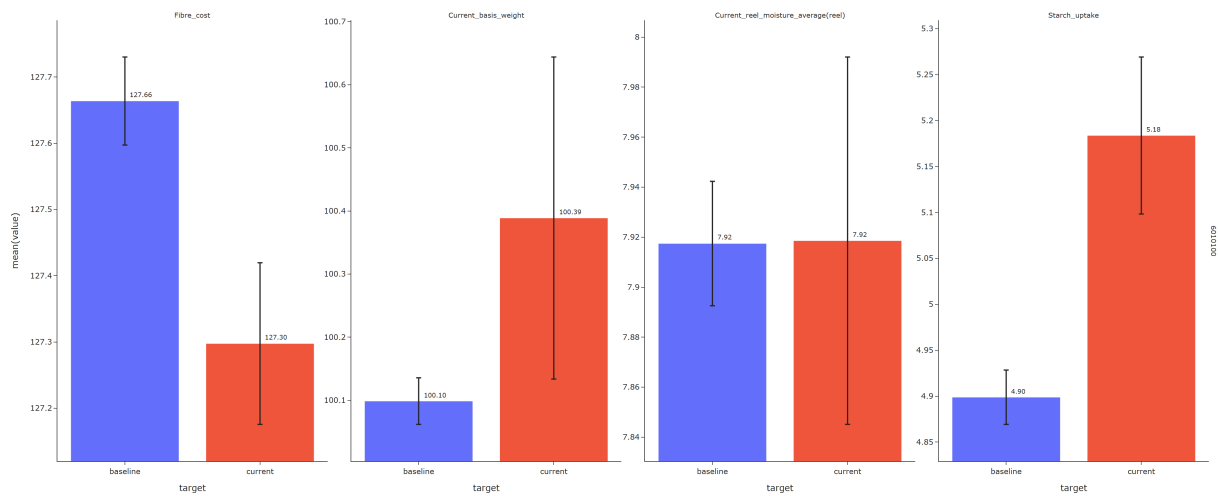


3.2 6010100

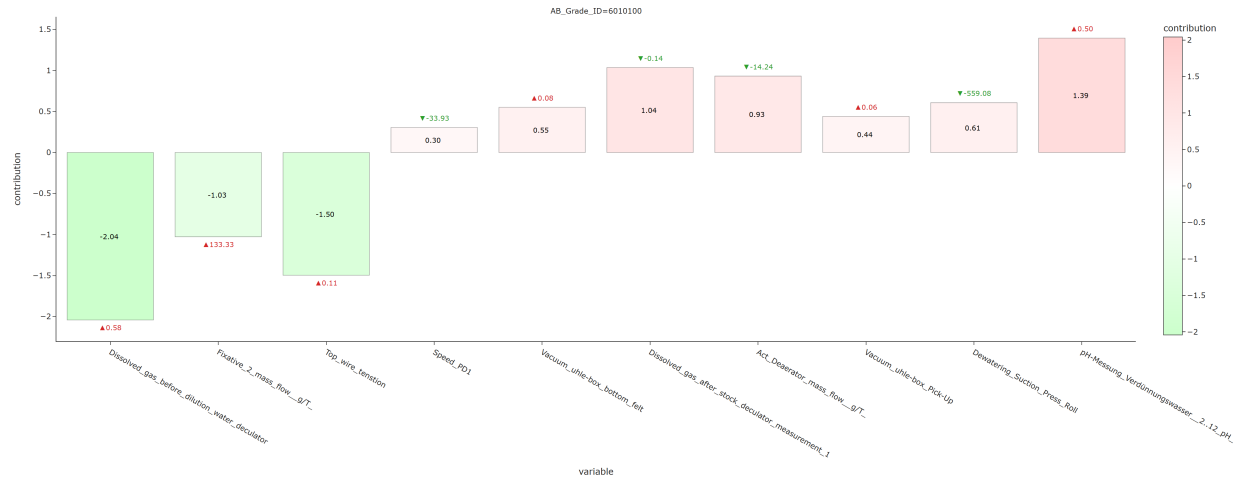
3.2.1 Fiber cost



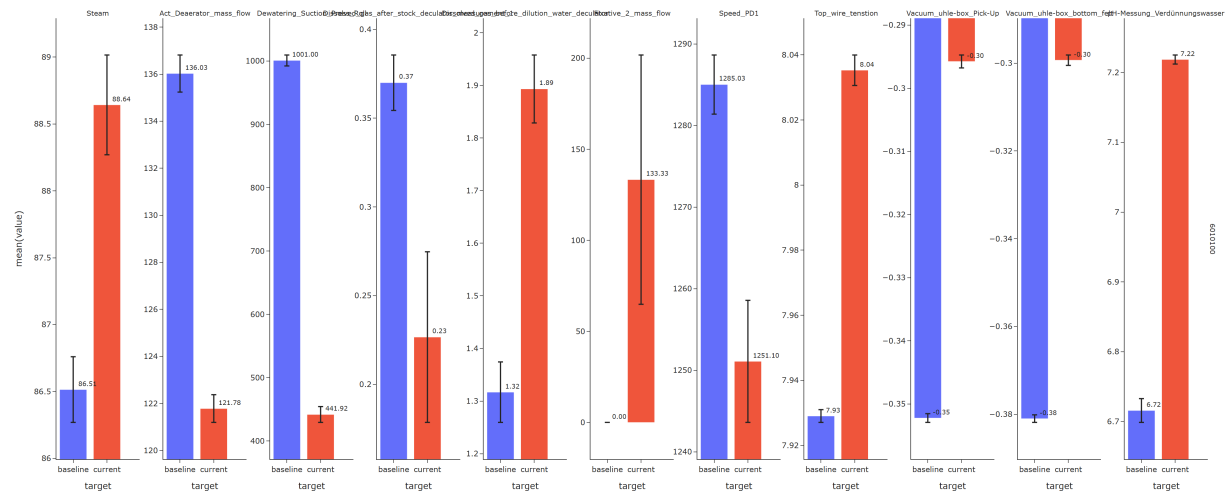
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Starch uptake g/m2 increased (+0.28)



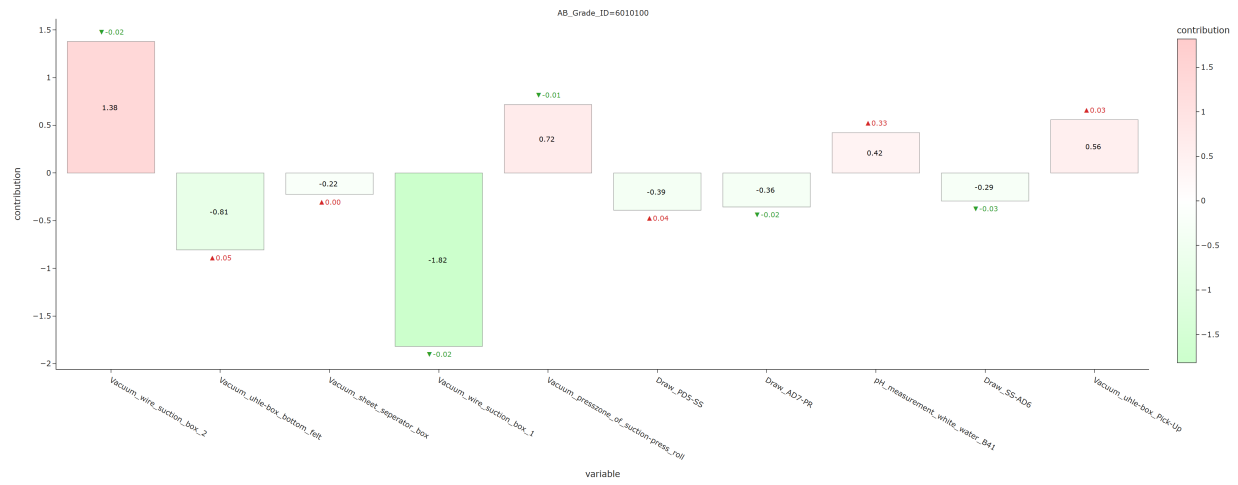
3.2.2 Steam cost



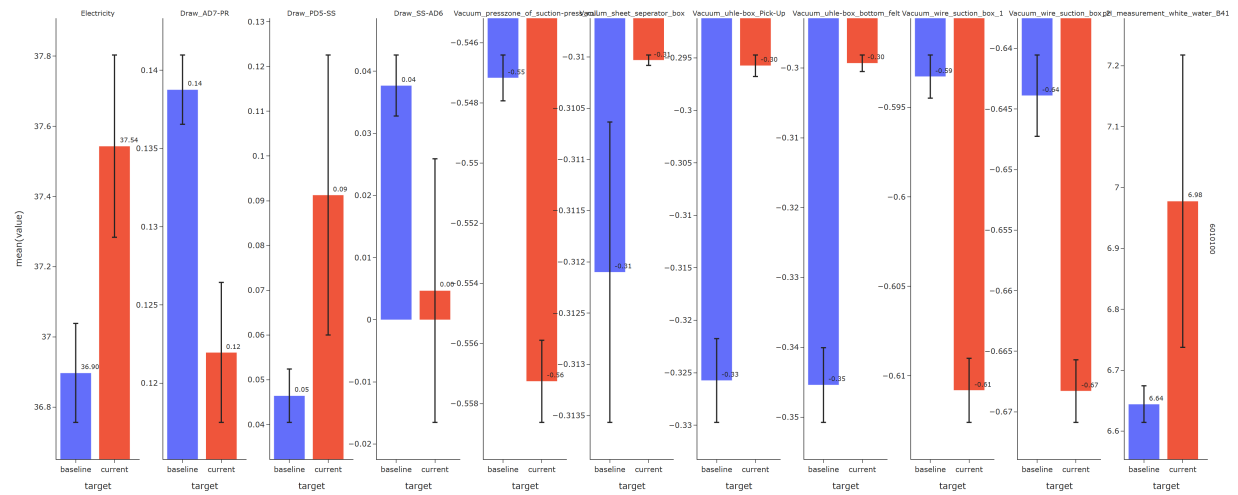
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Dissolved gas before dilution water decuator increased (+0.58) - Top wire tension increased (+0.11) - Fixative 2 mass flow g/T increased (+133.33) Cost-increasing drivers: - PH-Messung Verdünnungswasser 2.12 pH increased (+0.50) - Dissolved gas after stock decuator measurement 1 decreased (-0.14) - Act Deaeator mass flow g/T decreased (-14.24)



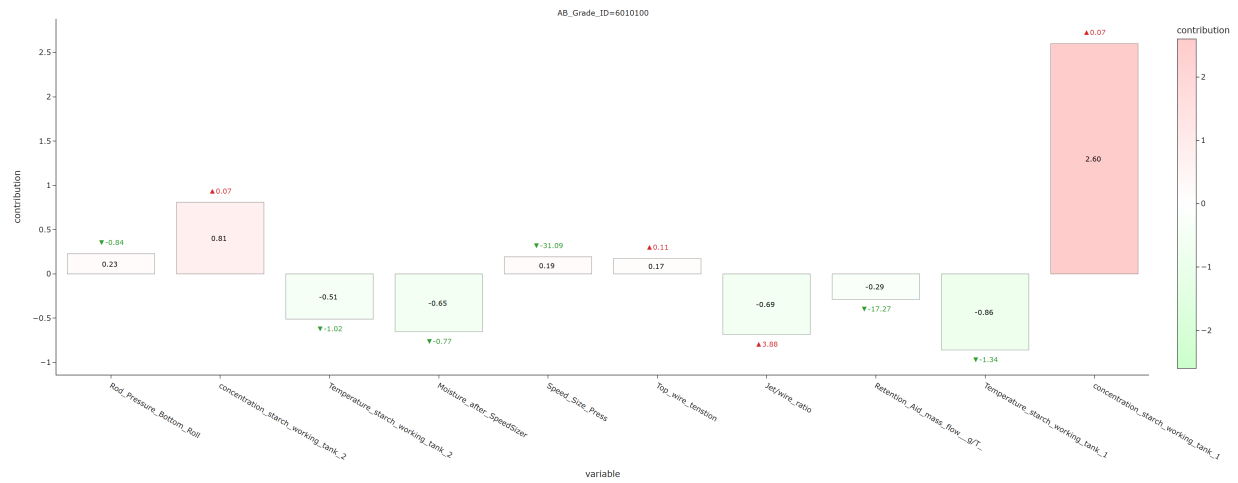
3.2.3 Electricity cost



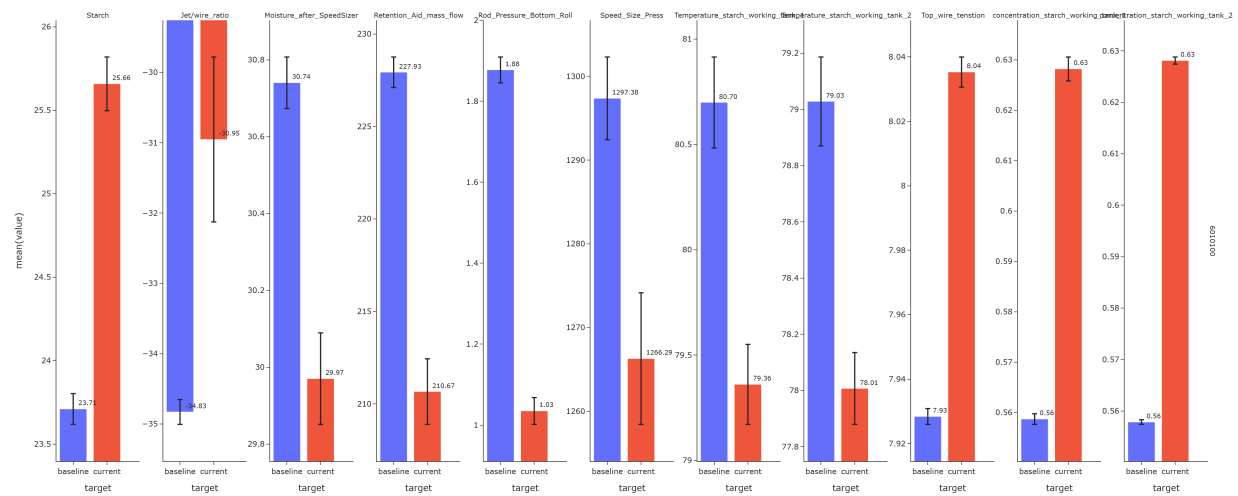
For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Vacuum wire suction box 1 decreased (-0.02) - Vacuum uhle-box bottom felt increased (+0.05) Cost-increasing drivers: - Vacuum wire suction box 2 decreased (-0.02)



3.2.4 Starch cost

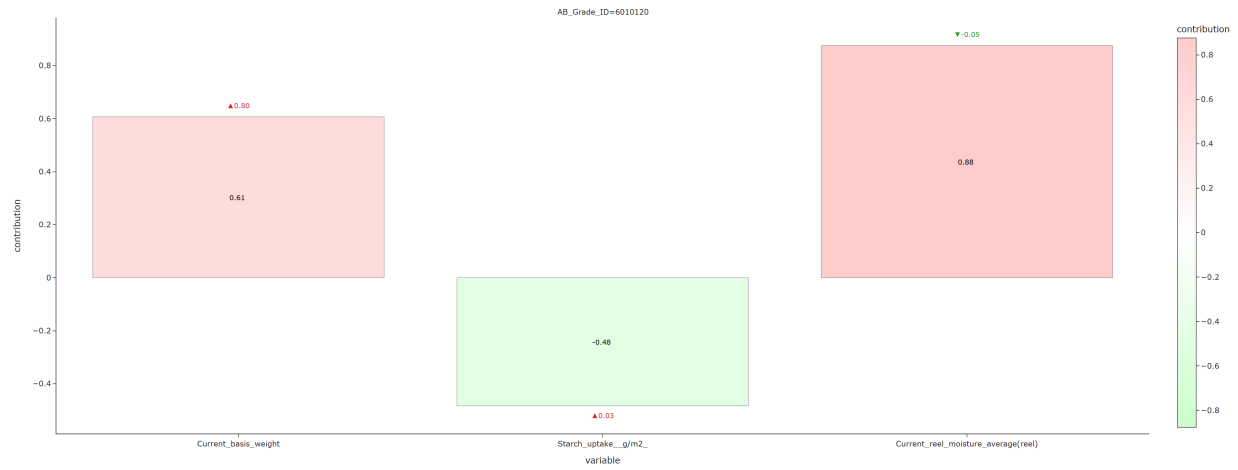


For grade 6010100, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Temperature starch working tank 1 decreased (-1.34) Cost-increasing drivers: - Concentration starch working tank 1 increased (+0.07) - Concentration starch working tank 2 increased (+0.07)

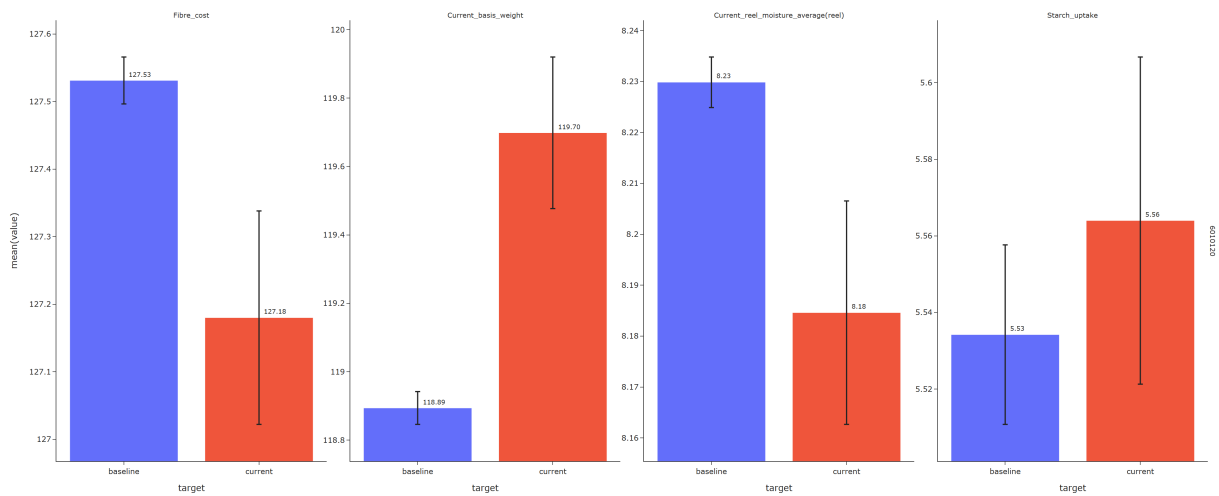


3.3 6010120

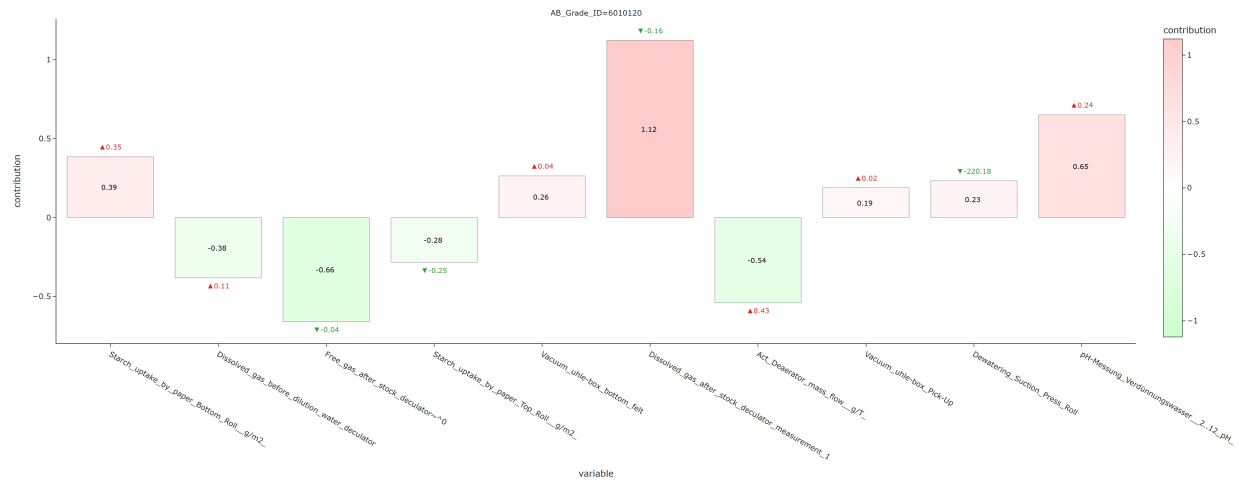
3.3.1 Fiber cost



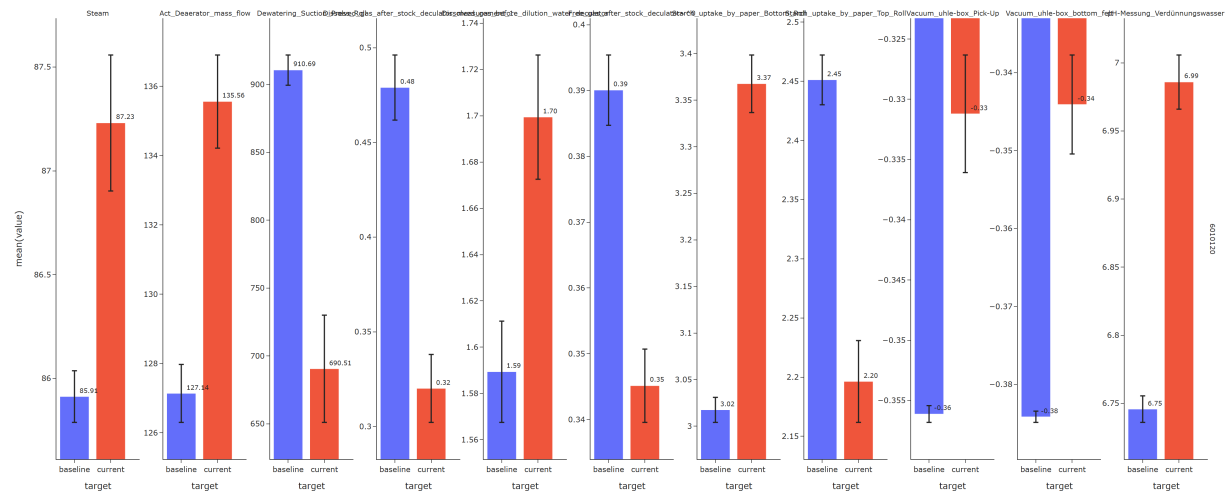
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-increasing drivers: - Current reel moisture average(reel) decreased (-0.05)



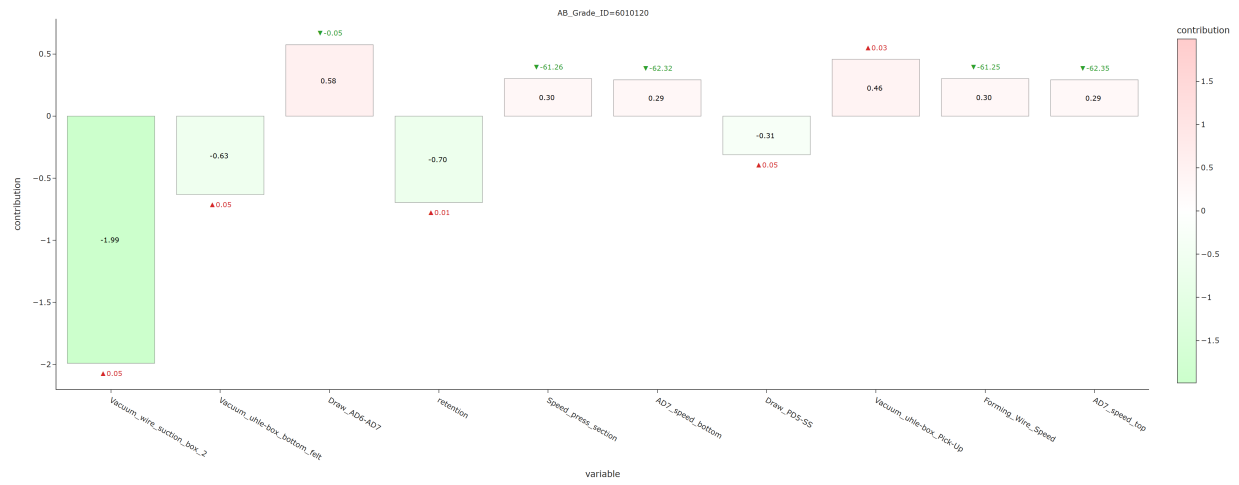
3.3.2 Steam cost



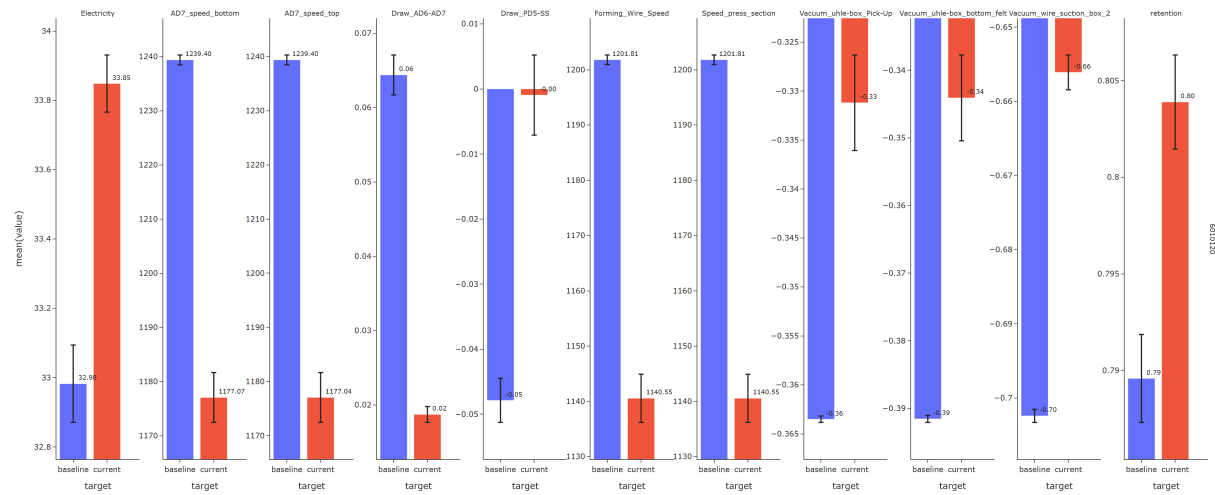
For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Free gas after stock decudator⁰ decreased (-0.04) - Act Deaerator mass flow g/T increased (+8.43) Cost-increasing drivers: - Dissolved gas after stock decudator measurement 1 decreased (-0.16) - PH-Messung Verdünnungswasser 2..12 pH increased (+0.24)



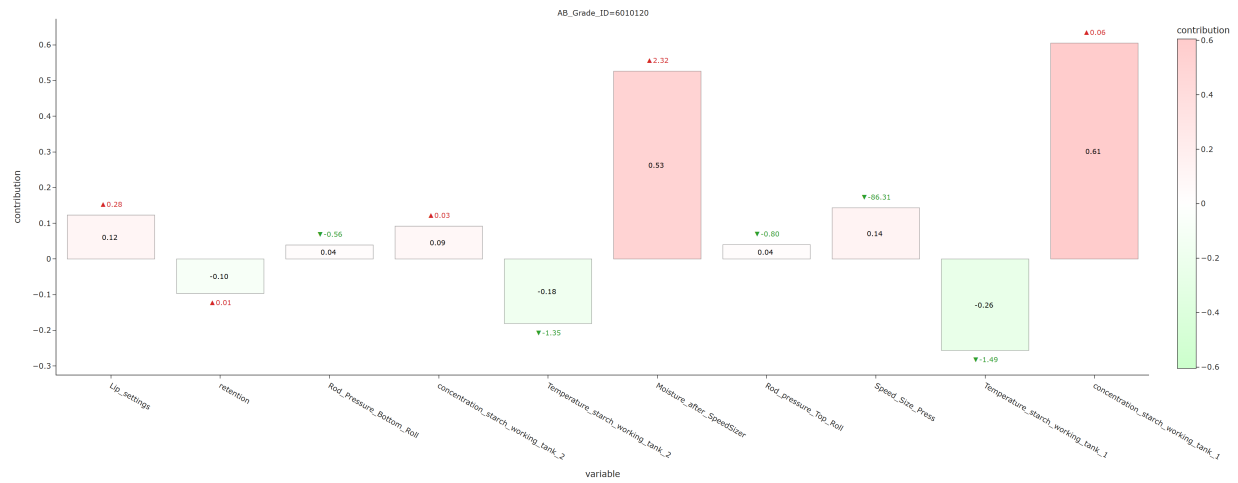
3.3.3 Electricity cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Vacuum wire suction box 2 increased (+0.05) - Retention increased (+0.01) - Vacuum uhle-box bottom felt increased (+0.05) Cost-increasing drivers: - Draw AD6-AD7 decreased (-0.05) - Vacuum uhle-box Pick-Up increased (+0.03)



3.3.4 Starch cost



For grade 6010120, the cost change is mainly explained by the following factors (top 20% by contribution): Cost-reducing drivers: - Temperature starch working tank 1 decreased (-1.49) Cost-increasing drivers: - Concentration starch working tank 1 increased (+0.06) - Moisture after Speed-Sizer increased (+2.32)

